Parkinson's Disease Gait Severity Classification Using Sensor-Based Data

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CERTIFICATE

This is to certify that the work embodied in this report entitled **Parkinson's Disease Gait Severity Classification Using Sensor-Based Data** has been satisfactorily completed by **Chaitya Dhandhara (24U020008) and Atul Joshi (24U020067).** It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering**, **Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Engineering during the academic year 2026-27.

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DECLARATION

We hereby declare that the following major project synopsis entitled **Parkinson's Disease Gait Severity Classification Using Sensor-Based Data** presented in the is the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**. It is an authentic documentation of our original work carried out under the able guidance of **Dr. Meenakshi Choudhary**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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AREA OF WORK

This project covers Machine Learning, Deep Learning, Time-Series Analysis, and Sensor-Based Gait Analysis for the classification of Parkinsonian gait severity. This work combines computational methods with movement data to investigate gait patterns linked to different levels of walking impairment in individuals with Parkinson's disease.

Features are linear and angular kinematic features measured via sensors that were recorded during a walking trials. These features are time-series data of multiple channels. The temporal series of these data gives enough features to enable computational analysis of the data. The clinical walking scores are used to define three severity classes.

It comprises the processes of data exploration, movement-data fusion, data quality evaluation, sequence mining, and feature scaling. Various machine learning and deep learning algorithms are tested and compared namely a one-dimensional Convolutional Neural Network (CNN1D) LSTM Bidirectional LSTM (BiLSTM), hybrid BiLSTM-CNN model, and Logistic Regression as a classical baseline model.

On top of that, the study explores how various settings in data-preparation impact the performance of classification. The examined models are assessed across accuracy, balanced accuracy, and macro F1-score to get a more comprehensive understanding of their performance across target classes.

The project operates at the intersection of Artificial Intelligence, Biomedical Data Analysis, and Computational Movement Analysis. Its goal is to investigate the use of sensor-based gait measurements for classifying gait severity and to provide a foundation for future research into reliable and generalizable gait-assessment methods. The system is designed for experimental analysis and does not replace professional clinical assessment.

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ABSTRACT

The Parkinson's disease (PD) is a progressive neurodegenerative disease with various clinical signs including motor impairments, such as changes in gait, movement, balance and walking. To assess the motor function of patients with Parkinson's disease and their ability to walk independently, gait features can serve as markers of impaired motor function. This work investigates the classification of the severity of Parkinsonian gait based on sensor data and machine learning methods.

Data for the study were collected from 26 subjects on- and off-medication for gait recordings (where available). The analysis workflow comprises: (1) an investigation of the available data, (2) the combination of recorded movements, (3) data quality checks and (4) processing of recorded sequences for computational analysis. The target classes for the three-class classification task were determined from the clinical walking scores for the recordings.

We compare a one-dimensional CNN (CNN1D), a Long Short-Term Memory network (LSTM), a Bidirectional LSTM (BiLSTM), a combined CNN1D-BiLSTM model and a Logistic Regression model as a classical baseline. We evaluate the models by means of classification metrics (accuracy, balanced accuracy, F1-score) and study the influence of different data-preparation configurations on the classification results.

The highest test accuracy found was 98.47%, the highest balanced accuracy was 97.89%, and the highest macro F1-score was 98.30% for all the configurations tested. However, the use of trial-wise data split means that there could be some test data from the same subject in other parts of the data set, hence the results should be viewed with caution as they might not generalize well to unseen subjects.

In general, the work provides the basis of a systematic approach for assessing gait severity by means of sensors and for comparing the performance of machine learning and deep learning approaches for the analysis of Parkinsonian gait. Future work is required to assess the robustness, generalizability and real world applicability of the computed results.

CHAPTER-1
INTRODUCTION

1.1 Background

Parkinson's disease (PD) is a slowly progressive neurological disorder characterized by disturbances of movement, coordination, and the performance of routine activities [2]. Typical motor disturbances in PD are postural disorder, muscular rigidity, akinesia/bradykinesia, and tremor. Impairments in gait is also a significant part of the disease as it can influence walking, balance, independence, and mobility [1].

Parkinson's disease patients walk with a shorter stride length, with increased shuffling of their walk, with decreased arm swing and with an altered walk rhythm. Some patients can also experience so-called 'freezing of gait' periods where they suddenly are unable to move while still wanting to walk. In Parkinson's disease patients, gait changes can vary greatly from person to person and even can vary from on- to off-medication within the same patient. Therefore, the analysis of movement is of great interest in order to gain more insight into the changes in motor function of the patient.

Clinical assessment of people with Parkinson's disease is done to assess the severity of their motor symptoms. The Movement Disorder Society-sponsored revision of the unified Parkinson's Disease rating Scale or (MDS-UPDRS) [3] was developed as a framework for assessing symptoms and the degree of motor impairment in Parkinson's disease. Although clinical ratings give insight into a person's condition, movement recordings obtained during walking trials reveal another means of assessing gait changes over time.

Developments in biomechanics and motion-capture allow for detailed measurement of human movement. The resulting features can be expressed as numbers and are recorded as a sequence of time steps, thus forming multichannel time-series data. Examples of public available datasets of gait of Parkinson's disease patients are given in [4].

A learning algorithm is used to learn relationships between the input measurements and the predefined labels. Deep learning models learn representations of the data and can be used for gait analysis [6], as convolutional networks can detect local structures in movement signals and sequence models such as recurrent networks can learn relationships between individual observations in a sequence.

The project aims to develop a computational framework for the classification of Parkinson's disease gait severity based on sensor-based gait analysis. The framework uses the recorded motion data as well as the clinical walking scores as labels for the classification task.

1.2 Motivation

Measuring of gait is not equal to measuring of several points. The gait is constant movement and therefore can be described by change of value from measurement to measurement. In addition, the sequence of measurements can be analyzed with respect to time-dependent structures in the changing measurements, which cannot be seen with single measurements. Deep learning time series classifiers are therefore suitable for such investigations [5].

On the other hand, analyzing of multichannel time-series data of 2 movement signals has its drawbacks too. There are several aspects to consider, such as recording length, range of numbers for features, possible errors in data like missing or even redundant values.

The choice of learning approach is of particular importance. For instance, convolutional networks can learn spatial patterns, while sequence models like recurrent networks can learn relations between observations of a sequence. For learning temporal structures in sequences of observations, bidirectional recurrent networks can learn a representation that is based on the information of both directions of a sequence. Hybrid approaches can incorporate complementary modeling approaches, while standard machine learning models can serve as good baselines.

However, each of these approaches has different strengths, and their performance will depend on the representation of the input, the characteristics of the dataset used for the evaluation, and the experimental setup [5]. Thus, in this work, we will perform a systematic comparison of several state-of-the-art models for the classification of Parkinson's disease gait severity.

The aim of the work described in this report is to establish a systematic approach for organizing gait recordings, and to investigate the use of different learning methods and their performance for classification purposes. Their applicability to the problem of gait-severity assessment will be explored in particular.

1.3 Overview of the Present Work

This project will focus on the categorization of the severity of Parkinsonian gait into three distinct classes using multichannel movement data. The data used for the project will be from gait recordings of 26 Parkinsonian subjects, with some subjects' on- and off-medication data available. The clinical data of the subjects' walking scores will be used as the target classes for the classification task [4].

The first step in the workflow of the project is to explore and check the data set. After this, linear and angular kinematic data are matched and incorporated into the data set. Subsequently, quality checks are performed on the data and the data is manipulated into the required sequential format for computational processing of the data in the subsequent steps.

The sequences thus produced are used for training and testing of five classification approaches, namely a 1-dimensional Convolutional Neural Network (CNN1D) [12], a Long Short-Term Memory network (LSTM)[13], a Bidirectional Long Short-Term Memory network (BiLSTM) [14] , a hybrid approach between the Bidirectional Long Short-Term Memory network and the Convolutional Neural Network (BiLSTM-CNN) and a traditional baseline approach, i.e. Logistic Regression (LR). All five approaches can be used for learning patterns in sequences of movement in order to classify them for specific targets, such as the severity of gait in Parkinson's disease [5].

For the evaluation of the classification performance of the three label categories, the accuracy, the balanced accuracy and the macro F1-score are used [7],[23]. Moreover, in the experiments on data-preparation, the influence of different configurations of the data-preparation steps on the classification performance is analyzed.

The dataset is split into training, validation and test sets at the trial level. It is worth noting that some subjects might appear in more than one partition even though some of the trials from the training set are omitted in the corresponding test set. Therefore, the test set provides a measure of performance for the current trial-wise split only and is not independent from the training set. Therefore, the performance of the models on independent individuals needs to be estimated taking into account the dependencies between the observations in the different partitions [8],[16].

This project is a first step to a standardized experiment setup for sensorbased gait-severity classification. The results of the project can be used to compare different approaches, to find out the weak points of the current setup and to make the first steps for future research.

1.4 Scope of the Work

The scope of this project is limited to processing and analysis of pre-recorded gait data. The focus of the work is on the comparison of several machine and deep learning approaches for the classification of Parkinsonian gait severity [4]. For this purpose the data are explored, combined and movement data are integrated into the data set. Quality of the data is analyzed and the data are prepared in a sequential manner for processing by the learning models [5]. The models are then trained on the data and their performance is evaluated.

This work does not intend to create a complete clinical diagnostic system, nor to replace clinical assessment or to clinically validate such assessment to work in a clinical environment.

Since this evaluation used trial-wise training, validation and test sets, results reported here are within the trial-wise partitions and therefore should not be generalized to testing on unseen subjects. To make a more fair comparison, subject-independent testing would have to be employed [8].

The following chapters describe the research problem, related works, the methodology, the algorithm, the system pipeline, the implementation technologies and the results of the experimental part of the project.

CHAPTER-2
LITERATURE REVIEW

2.1 Introduction

With the growing use of computation to analyze the behavior of Parkinson's disease (PD) patients, gait analysis has become a prominent method. Sensors and motion capture have been used to gather movement data, and machine learning methods have been applied to identify patterns of movement indicative of motor impairment. These computational methods complement the traditional clinical assessment, providing quantitative measures of movement [6], [9].

Work in the area of representation of the gait data has considered various approaches. These can be categorized as either hand-crafted gait descriptors or learned features from deep learning. The representation of the data, the architecture of the system and the evaluation protocol all have an influence on the success of the classification system and its interpretation [5].

This chapter starts with a background of sensor-based gait analysis and then provides a review of state-of-the-art machine learning/deep learning methods, temporal modeling approaches and the computational gait-classification system evaluation problem. After the literature review, the chapter ends with the scope of this project.

2.2 Sensor-Based Gait Analysis in Parkinson's Disease

Gait analysis can be used to assess the patient’s disease as well as the degree of motor impairment. Movement measurement data such as gait features can be stored as a sequence of numbers and analyzed in order to find out whether there are any changes in the gait features. These quantitative gait parameters can be used as additional clinical measurement of movement disorders [2], [10].

A systematic review on the use of machine learning and wearable sensors for the diagnosis of PD has been presented by Di Biase et al. [9]. After reviewing the different types of sensor based measurements, including those related to gait, the authors discuss the use of computational algorithms as objective measurement tools. Challenges to the implementation of such methods in real-world scenarios are also presented.

Another review has focused on the use of deep learning for the analysis of movement in the Parkinson’s disease population. This review has covered the range of different sensors that have been used to measure movement, and the different types of learning that have been used

to analyze the movement data. The authors also highlight the increasing use of computational gait analysis and the need for appropriate evaluation and interpretation of the results [6].

The data to be used in this work was characterized by Shida et al. (2012) in their public dataset of full body gait kinematics and kinetics of 26 patients with Parkinson's disease (PD) recorded in both ON- and OFF- medication states. This data includes movement recordings as well as contextual clinical data for further analysis of gait [4].

As for the quantitative movement analysis itself, the recording environment, the measurement setup and the subjects to be analyzed can vary considerably. All these factors are taken into account in the following report.

2.3 Machine Learning and Deep Learning for Gait Classification.

The application of machine learning algorithms for feature learning on Parkinson's disease movement has also been explored. A systematic review was conducted by Sharma et al. (2018) to study the application of machine learning on Parkinson's disease using the gait features [11].

These learned representations can be even more powerful for gait analysis than for other applications of time-series classification, since the movement recordings consist of observations at different time points.

However, a more pertinent aspect of the use of deep learning for the classification of gait signals is that the movement recordings consist of observations of features of movement of the human body, stratified over time. Therefore, deep learning methods for time-series classification can be used as a starting point for the investigation of the sorting of such sequences [5].

A 1-D convolutional neural network (CNN) was proposed by El Maachi, Bilodeau, and Bouachir (2015) for the detection of Parkinson's disease and its severity quantification from gait signals [12]. The system used signals from force sensors attached to the bottom of the foot and automatically learned features from the data without the need of a separate feature-extraction step. This work demonstrates how convolutional learning can be used for movement classification as well as severity quantification.

Convolutional models learn local structures of structured sequential data such as gait recordings.

Moreover, the relationships of longer intervals of a walking sequence may be more relevant for the classification of gait signals than the convolutional models can learn from local structures.

Hence, besides the CNN, also recurrent neural networks as well as hybrid models are considered in this work.
Comparison of different architectures for gait signal classification, such as CNN, RNN, BiRNN and hybrid models, as well as comparison with a simple Logistic Regression model for better baseline comparison will be presented within a single uniform experiment.
The results will be compared within a uniform experimental environment.

2.4 Temporal and Hybrid Deep Learning Approaches

The data of gait recordings are represented as time-series data. In order to model the associations between adjacent observations of features of motion, recurrent neural networks can be used. For learning dependencies within sequential data, Long Short-Term Memory networks (LSTMs) were introduced [13].

As with the previous approach, the bidirectional recurrent neural networks take the input sequence and process it in both directions, forwards and backwards, thus utilizing information from past and future observations [14]. Therefore, the Bidirectional LSTM models are also very well suited for analyzing the temporal structure of the gait recordings.

Quantitative assessment of Parkinson's disease with a deep neural network. Xia et al. (2018) proposed a dual-modal deep learning network for quantitative assessment of features of Parkinson's disease [15]. The network consists of convolutional layers for processing and an attention-enhanced LSTM network for classification using ground reaction force signals corresponding to gait from left and right sides. This study showed that the use of convolutional and recurrent modules can provide useful gait pattern analysis for understanding the disease severity of Parkinson's disease .

This work evaluates the performance of simple hybrid architectures (like a CNN with a subsequent RNN) and more complex architectures on classification tasks with local features and temporal dependencies. Simple models are not always outperformed by more complex ones, but can be outscored by hybrid models that combine the advantages of both .

We are interested in studying the relationship between traditional machine learning and alternative deep learning models in the context of our classification task [6]. Hence, we compare the CNN1D, the LSTM, the BiLSTM, the hybrid architecture BiLSTM-CNN and the simple Logistic Regression model in the same experiment.

2.5 Research Challenges and Identified Gap

The main challenges in developing and evaluating approaches for the classification of Parkinson’s disease (PD) gait using computational approaches to gait analysis are data variability, representation of the movement signals, model selection, and the results from experiments.

Gait recording inconsistencies: Walking sequences may vary in length, in the way gait is performed and in recording conditions. Therefore, before a classifier is applied to sequential data, a representative must be generated[5].

Multichannel data analysis: The signals recorded from the movement of Parkinson's disease patients can be represented as multiple channels. These can be used in combination to account for data quality, feature coherence and the interrelations between the signals. This is also the case for the publicly available datasets of Parkinson's disease patients' gait. However, these datasets differ considerably in terms of the kinematic and kinetic measurements that are recorded [4].

The choice of learning algorithm may highlight different features of the input data and thus need to be evaluated on a per-case basis, as there is no general rule for this.

Another important point to be considered is that models, which are tested on test sets consisting of subjects that were also in the training set, can suffer from data leakage. Thus, the way of constructing the training, validation and test sets is of high relevance. In studies on the validation strategy as well as on machine-learning data leakage, the dependencies between the observations have to be considered [8], [16].

Comparability of the results reported: It is possible to compare different studies of gait analysis of Parkinson's disease, even if different data sets, clinical targets, measurement systems, and evaluation approaches are used in the studies. This is demonstrated in a systematic review of the field [6], [9].

Evaluation of the classifier performance: The performance of the classifiers can be evaluated using several measures, such as the accuracy, the balanced accuracy, and the macro F1-score [7], [23].

These points highlight the need for a systematic, transparent workflow to the analysis of such data, from data preparation to model evaluation and the results reported in the study. All steps of the experiment need to be clearly described to ensure that the results can be properly understood.

2.6 Relevance to the Present Work

This collection of papers contains some examples of quantitative gait analysis of PD patients as well as some of the current literature that uses machine learning and deep learning for the analysis of quantitative gait data from the same PD patients. The results of quantitative gait analysis are highly dependent on the quality of the data as well as the test conditions [6], [9].

This work explores the three-class gait-severity classification based on the measurements mentioned above. The recordings of 26 subjects with Parkinson's disease, in ON- and OFF-medication conditions (if available), were used [4]. The target labels for the classification were chosen from the clinical walking scores.

The work flow of this project consists of the following steps: dataset exploration, movement-data integration, construction of sequential input for models, comparison of performance of different models (CNN1D, LSTM, BiLSTM, BiLSTM-CNN and Logistic Regression) with the same classification task on the same data set using the same set of classification metrics for comparison between traditional machine learning and alternative deep learning models for temporal sequences [6].

In addition to the aforementioned problems, the evaluation of learning methods also requires careful design of experiments and transparent reporting. In this paper, we focus on the comparison of several learning methods for Parkinson's disease gait severity classification using the aforementioned dataset and experimental setup. We employ a trial-wise split of the data, which means that recordings of the same subjects can be in different splits. Therefore, the results reported in this paper do not necessarily generalize to new subjects. Additional experiments are required to demonstrate subject-independent generalization.

The current literature supports the sensor-based gait classification for Parkinsonian gait severity. However, other remaining method-related problems have to be experimentally investigated and reported in detail. This project can be considered as a contribution to the above-mentioned problems by comparing different learning methods on the Parkinsonian gait severity classification within the framework of this project.

CHAPTER-3
PROBLEM DEFINITION AND OBJECTIVES

3.1 Problem Definition

Parkinson's disease may impact walking and movement throughout the body. Examining gait recordings enables us to study movement features that can be associated with various levels of gait impairment. Although, simply observing multichannel movement recordings is difficult to interprets in real-time because gait is a time series of many measurements. Earlier studies have explored quantitative gait measurements as an indicator of movement deficits related to Parkinson's disease [2].

Machine learning and deep learning algorithms are computational algorithms that learn the association between recorded movement patterns and predefined clinical classes. However, constructing a robust classification algorithm necessitates an explicit consideration for issues such as data quality, movement-sequence representation, machine learning algorithms, and evaluation methods [5].

The task of this project is to classify the severity of gait disorders, calculated based on sensor readings, into three levels, by applying supervised learning algorithms to sensor readings and selected methods for artificial intelligence. This work utilizes gait data of 26 subjects and ON- and OFF-medication status are available for selected subjects in this data collection, with clinical walking scores being input labels (classification labels) based on clinical information for this gait dataset [3], [4].

The core question of this project is whether motion sequences of a particular gait severity can be recognized across all gait severity levels and whether a learning algorithm can improve upon this, given the same experimental conditions. The effect of the dataset and evaluation strategy on the interpretation of classification results is also presented.

This project is an experimental study of computational gait classification. It is not meant to substitute clinical assessment or define a clinically validated diagnostic system.

3.2 Aim of the Project

This project aims to create and assess machine learning and deep learning classifiers to determine if a person's class of Parkinsonian gait severity is accurately three levels using kinematic data (recorded video data) that has been captured. The project will compare various models to test how well a model can learn movement for classes.

3.3 Objectives

The main objectives of the project are:

Analyze database recordings, subject information, medications used by participants and clinical labels of course.

The uniform unit of measure for linear and angular kinematic variables is defined and potential data quality issues are addressed before model training.

CNN1D, LSTM, BiLSTM, hybrid BiLSTM-CNN and Logistic Regression are implemented as alternative approaches for learning from sequential data [12].

Using accuracy, balanced accuracy and macro F1-score we compare the different models' classification performance [7].

The effect of different data-prep configurations on the classification performance of the chosen models.

Finally, we discuss the positive and negative aspects of our results. In particular, we will discuss the implications of the trial-wise partition, where the same subject can be present in different data partitions. We will also discuss how the design of the evaluation can affect the conclusions that can be drawn regarding the generalizability of the results to independent people [8].

Show improvements, i.e. testing on subject-independent partitions and perform additional validation to determine the generalizability of the approach.

3.4 Scope and Limitations

This project investigates the computational analysis of the recorded gaits, the definition of multichannel kinematic sequences for the various activities, the training of the classifiers and the comparison of the different classifiers for the three-class classification problem.

Each trial was split into separate training, validation and test sets [8]. Since the same subject may have participated in more than one trial, the results do not provide an estimate of how well the model will generalize to new subjects. Rather, they provide an indication of how well the model performs given the chosen experimental design. In order to address the dependencies between the samples, cross-validation and machine-learning data leakage literature recommends addressing dependencies between samples [16].

However, the small population makes general conclusions to larger populations impossible. Additional trials with more recordings and independent test partitions for subject-independent evaluations would be required for more reliable results.

Therefore, the results presented here are of an experimental nature and do not yet indicate clinical applicability. This project is far from being a clinical diagnostic system and in no way is it intended to replace human judgment. Additional validation is required in order to assess potential clinical applications of the proposed approach.

CHAPTER-4
PROPOSED METHODOLOGY AND WORK DESCRIPTION

4.1 Overview of the approach

This work provides a detailed description of the proposed flow to classify the severity of Parkinsonian gait based on recorded human movement data. The flow is composed of four main steps, which are dataset exploration and validation, dataset preparation, classification models development, and experimental evaluation.

We used standard classification performance measures for evaluation of several machine learning and deep learning classification methods on the linear and angular kinematic signals of recorded human movement data, which were transformed into multichannel time-series data [6]. Time-series classification by means of deep learning has been extensively studied and is currently the benchmark to test other sequential data classification methods [5].

The main stages of the proposed method are as follows:

1. Exploring and validating the available data.

2. Recognition and coupling of matching motion values.

3. Assessing the data quality of the sequential inputs and possibly processing them in this stage.

4. Machine Learning / Deep Learning Models.

5. Model training, validation, and testing of the models selected.

6. Comparison and analysis of the experimental results.

4.2 Dataset and Input Representation

The recordings of 26 subjects, of which some were tested in both ON- and OFF-medication state, were used for the dataset. The dataset consists of the kinematic and kinetic data as well as the clinical data of the participants. The source dataset for this study contains walking-related kinetic and kinematic data of Parkinson's disease gait that can be used for research purposes [4].

The input data for the model are the linear and angular kinematic variables of the different movement components. These variables were then combined to form a multichannel time series of the movement's temporal evolution. The time series was kept in the sequential order of the

original data in order to retain the characteristic features of the movement, which are only present when several observations are considered.

The categories for this dataset are described with clinical walking scores. The model has to learn to map a given movement sequence to the corresponding severity category. The MDS-UPDRS measures motor experiences and motor impairments of patients with Parkinson's disease [3].

The data structure was analyzed for the data set exploration in the first step. For this purpose, it was checked for each of the recordings if there are valid target labels and if the input measurement values can be used in combination with the target labels.

4.3 Dataset Exploration and Data Integration

We first analyze the files of the subjects, the medications they take, the movements they perform and the clinical metadata for each of them. The distribution of dataset is shown in Table 1. We then have to find out how the recordings of each subject were organized and how to link the files of the walking trials of each subject to their recordings.

In the second step of the analysis the linear and angular kinematic features of the trials are paired with the corresponding trial data using the label and trial information from the dataset. The result is a combined data representation for each trial which still refers to the original label of that trial.

Checking the integrity of the data to find out errors in the data that can cause problems in the rest of the processing of the data. Checking for the existence of the recordings, checking the sequence features, checking the training data for consistency.

The structured set of gait recordings and corresponding target labels are prepared for the subsequent stage.

Table 1: Class wise Distribution

Class	Trials	Patients-ON Medication	Patients-OFF Medication
0	381	14	6
1	295	6	9
2	205	5	8

4.4 Data Preparation

The raw actions' data can contain missing values, sequences of different lengths and numerical values. This must be considered when using machine learning for processing sequential data [5].

All missing values were evaluated and possible data quality problems were handled. For each metric the data were transformed into a sequence of fixed length of uniformly processed input variables while the link to the corresponding clinical target label was maintained.

We investigated the impact of six different data-preparation approaches on the accuracy of the models, without making any assumptions about the individual models.

The input sequences are represented as fixed-length multichannel gait sequences, each with 751 time-steps and 180 kinematic features per time-step (i.e. frame and time fields are not considered). These sequences are then used as input for the models to be trained and evaluated.

4.5 Model Development

We evaluate the performance of four different deep learning models as well as the performance of a simple machine learning model that serves as our baseline. These five models are different ways of trying to extract patterns from the input data and mapping them to the different classes of the output variable.

4.5.1 One-Dimensional Convolutional Neural Network

CNN1D tries to find local structures within the sequences of observations of movements. Convolutional layers can detect structures within adjacent observations. This can help to classify between different classes of movements.The architecture of model is shown in Fig. 1.

Local features can be very effectively used for pattern recognition by convolutional neural networks [22]. For analysis of multiple channel gait sequences, CNN1D is tested as a method.

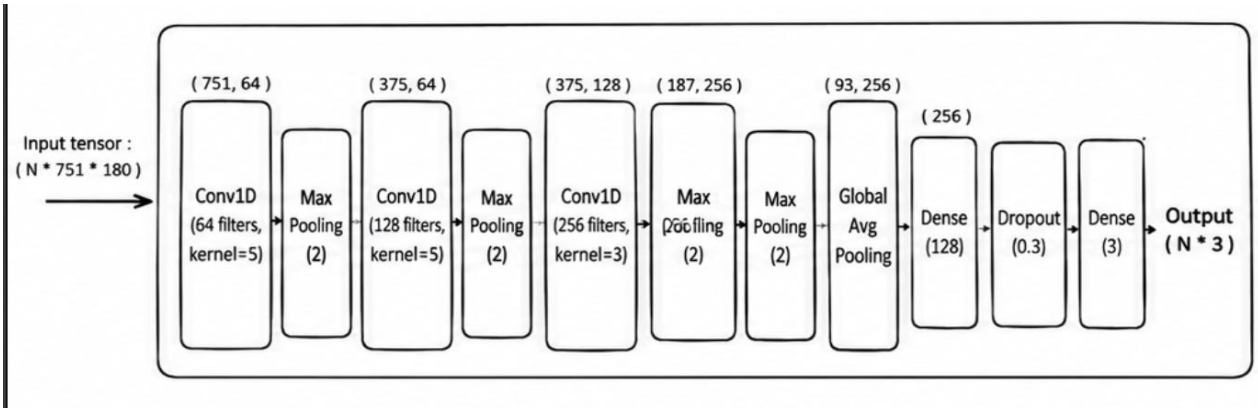


Fig. 1: CNN Architecture

4.5.2 Long Short-Term Memory Network

A recurrent neural network called LSTM (Long Short-Term Memory) network can learn dependencies within sequential data. It can store data over many time steps and thus recognize time-dependencies of the gait-severity categories [13]. The architecture of model is shown in Fig. 2.

The model is trained to examine if the recurrent sequence learning can find any structures within the prepared motion recordings..

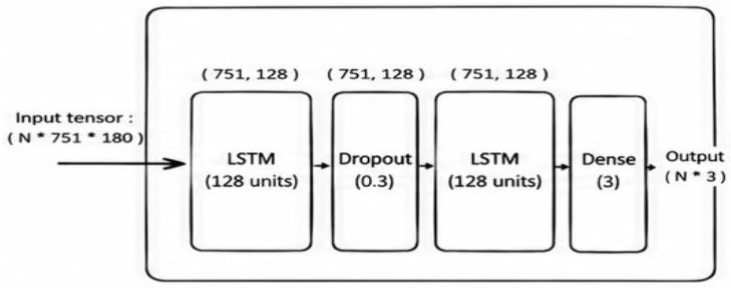


Fig. 2: LSTM Architecture

4.5.3 Bidirectional Long Short-Term Memory Network

For the BiLSTM we use hidden units with the same dimension as the word embeddings of the model. The architecture of model is shown in Fig. 3. The BiLSTM is a bidirectional temporal model. It can use information from past and future time steps to represent the current time step [14].

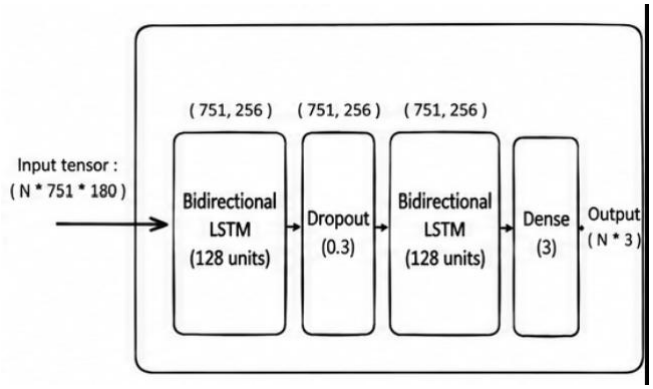


Fig. 3: BiLSTM Architecture

4.5.4 Hybrid BiLSTM-CNN Model

The RNN part of the hybrid model is used to extract the temporal features of the input sequences, whereas the CNN part of the hybrid model is used to extract the local features of the input representation. The architecture of model is shown in Fig. 4.

In the related work [15] a fused convolutional and recurrent neural network was applied to analyze the gait of Parkinson's disease patients. In this work the complementary modeling approach of the fused network is investigated to improve the classification performance.

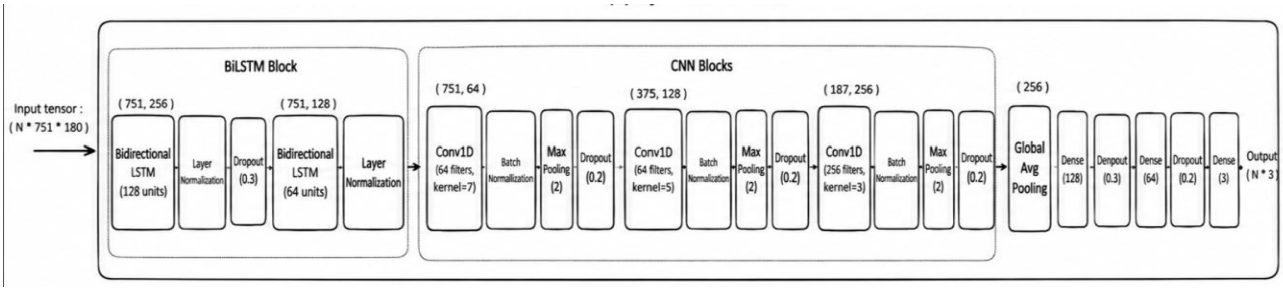


Fig. 4: BiLSTM-CNN Architecture

4.5.5 Logistic Regression Baseline

Logistic Regression - Included as a traditional machine learning baseline, it serves as a comparison point for assessing the performance of deep learning models.

The performance of different models can be compared on the same classification task by using the baseline model [4].

4.6 Training, Validation, and Testing

All of the generated trials were then split into three separate partitions for the purpose of the trials' pre-processed gait sequences and their respective labels being learned by the various models' parameters. These partitions consisted of 70% of the generated trials for training purposes, 15% for validation purposes, and 15% for testing purposes. For the third of the experiments, for example, there were 619 training trials, 131 validation trials, and 131 testing trials.

The generated trials are split into three sets of data for the purpose of learning the model parameters from the pre-processed gait sequences and their respective labels.

The validation set is used during training to monitor performance of the model and to select the best model and/or checkpoint.

This set is used to test the models that were selected on the trials that were not used for training of the respective models. The predictions for these trials are compared with the original target labels to determine the metrics.

One limitation is the fact that the split into training, validation and test set was performed at the level of the trials and not the subjects [16]. Thus, similar subjects might have been included in the training, validation and test set, respectively. This does not provide an independent evaluation of the generalization performance on unseen subjects. Thus, for evaluation

purposes, the correlation between the observations must be taken into account when evaluating the performance on independent subjects [8].

4.7 Evaluation Metrics

We use multiple classification metrics to assess the performance of the three models on the three targets. These are more informative than a single classification metric.

Accuracy: The average of correct classifications for all observations of a model. The accuracy does not provide any information on how well a model performs on individual classes.

Balanced accuracy: The average of the recall values of the individual classes with equal weight to each of them. For multiclass problems the balanced accuracy is often more informative than the accuracy, see [23].

Macro F1-score: This metric calculates the F1-score for each class and averages them up. Each class gets the same weight in the average. Thus, no class dominates the average more than any other class, even if that class is much more frequent than other classes in the data.

All of the above classification metrics were used for model comparison in order to select promising models for further investigation. All results and corresponding conclusions presented in the following are restricted to the data of the trial-wise evaluation of this work.

4.8 Overall Workflow

The complete methodology is outlined below:

Recording identification and data integration -> Data-quality evaluation -> Sequence preparation -> Model development and validation -> Training and validation -> Test-set analysis -> Comparison

The goal of the analysis is to classify the data recorded by the recorders. All information from the recordings can be retrieved by analyzing the recorded data. The data preparation step of the general methodology for sensor based Parkinsonian gait-severity classification transforms the recorded data into one input format.

The results of the different models are compared and limitations are addressed in the subsequent steps of the general methodology for sensor based Parkinsonian gait-severity classification.

CHAPTER-5
PROPOSED ALGORITHMS

5.1 Algorithm Overview

The algorithm for Parkinson's disease gait severity classification presented in this paper processes the multichannel human movement signals. Detailed structure of used signals and models for training, testing, prediction and evaluation of results are presented in this chapter.

The five different models for Parkinsonian gait classification presented in this chapter can learn the complex, sequential relationships within the sequential data to make accurate predictions. As already mentioned in [13] and [14], the LSTM and BiLSTM models are especially suited for analyzing temporal relationships within the sequential data.

All models can be compared in the same framework. For each model individual approaches for training can be used.

The proposed algorithm can be used to process gait-sequence data of pre-processed trials and the corresponding clinical target labels. The gait-sequence data used in the algorithm consists of 751 time points per sequence and 180 features per sequence (the 180 kinematic features). The output of the algorithm are the predicted gait-severity scores for the used model and the set-up tested.

5.2 Proposed Algorithm

Algorithm: Multichannel Gait-Severity Classification

Input:
- Gait recordings from the available subjects.
- Corresponding clinical walking-score labels.
- Selected data-preparation configurations.
- A collection of classification models.

Output:
- Predicted gait-severity categories.
- Accuracy, balanced accuracy, and macro F1-score.
- Comparison of experimental results between models and configurations used.

Procedure:

Step 1: Initialise the dataset.

Load the gait recordings as well as the clinical data of the subjects.

Step 2: Explore and verify the data.

For each of the clinical subjects verify that there are corresponding gait recordings, trials, movement measurements, and labels of the targets.

Step 3: Integrate movement data.

Combine the linear and angular kinematic data of the same trial into one representation.

Step 4: Assess data quality.

Check for missing values, other errors and other possible defects in the input recordings that can affect the analysis.

Step 5: Prepare the input sequences.

Prepare input sequences for each of the selected data-preparation methods and the corresponding model. The multichannel data are processed consistently for the sequences of different channels and are associated with the corresponding target label.

Step 6: Create data partitions.

The trials are split into three sets for the analysis: train, validation and test. This split is done trial-wise with about 70%, 15% and 15% respectively.

Step 7: Select a model and configuration.

For each of the data-preparation methods, select the corresponding model from the list of models given below for the corresponding configuration.

- CNN1D
- LSTM
- BiLSTM
- BiLSTM-CNN
- Logistic Regression

Step 8. Train the Model.

Train the model on the training set of data. If the model is trainable, map the sequences of input data to the classes of target data using the training procedure specified for the model.

Step 9: Validate the model.

The model with the best performance on the validation set is chosen (checkpoint selection if available).

Step 10: Generate test predictions.

The severity of the test set is predicted using the selected checkpoint of the model of step 9.

Step 11: Evaluate the predictions.

The severity predicted for the test set is compared to the true severity of the test set. For this purpose, several evaluation metrics are used, i.e. accuracy, balanced accuracy, and macro F1-score [7], [23].

Step 12: Record experimental results.

Store the model ID, evaluation configuration, test predictions, and metrics for comparison in Step 14.

Step 13: Reassess

Now the other models and configurations are trained, validated and tested in the same way as described above.

Step 14: Compare the results.

Look at the recorded data for each performance measure and find the maximum value for each measure. Compare the different approaches that were evaluated.

Step 15: Interpret the findings.

The findings from the comparison of the results are discussed with reference to the data set used for the evaluation and the evaluation configuration.

5.3 Algorithm Summary

The first complete algorithm for translating measured gait data into severity categories and verifying the classification performance of the algorithm is presented in this study. The performance of the algorithm is tested within a single experiment where the measured data are verified, the input variables are processed sequentially for the classification algorithm, the algorithm is trained and tested, and the algorithm is evaluated by means of quantitative measures.
The results of the experiments are considered as experimental results rather than as clinical results since the framework was not tested on novel test subjects in a trial-wise fashion.

CHAPTER-6
PROPOSED FLOWCHART / BLOCK DIAGRAM / DFD

6.1 Introduction

Each of the stages in the pipeline of the recorded gait measurements for predicting the gait-severity classes are outlined in this chapter. Each of the stages of the pipeline in turn provide the input for the next stage in the pipeline of the recorded gait measurements for predicting the gait-severity classes.

The workflow of the proposed system starts with data exploration and movement-data integration, followed by data-quality validation and sequence preparation. The prepared trials are then used for model training, validation, testing and comparative analysis of different learning strategies in a standardized experimental setting.

6.2 Proposed Block Diagram

The main components of the system and the data flow between them are shown in the flow diagram of Fig. 5. The input to the main system is the raw gait recordings of the subjects and the clinical information of these subjects.

The kinematic and kinetic features of gait are extracted from the source dataset of patients with Parkinson's disease. The target measures for the classification of motor function are defined by the clinical scales of the patients [3], [4].

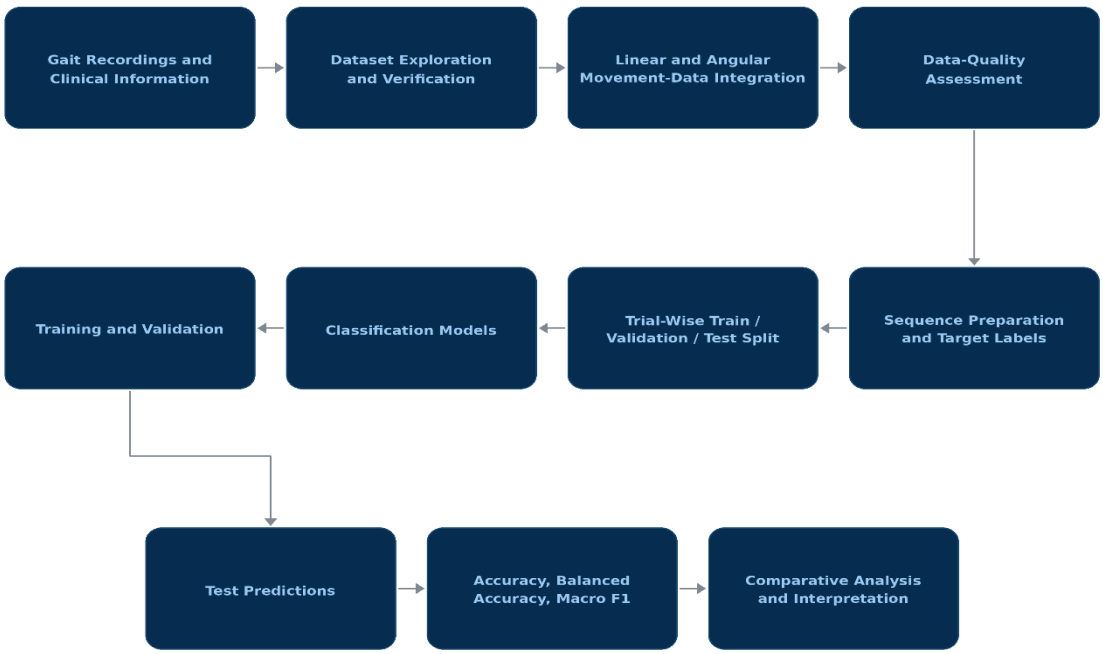


Fig. 5: Block diagram of the proposed gait-severity classification system

6.3 Description of the System Components

The input data of the system consists of linear and angular kinematic data of the gait of given subjects and the clinical data of these subjects. The clinical data consists of walking scores of the subjects and the target classes of the system are defined by these scores as given in [3], [4].

The first step of the process is the exploration of the subject's recordings, the walking trials, the recording conditions and the measurements. The movement data are then brought together to form a single image of each gait trial.

The missing values, inconsistencies and other errors that are contained in the assembled data are identified and documented in this step.

The movement measurements must be transformed into a format that can be processed by the models. It is also important that the sequence to true label association is preserved so that the model predictions can be compared with the true labels.

The trial preparation process produces the prepared data for the training, validation and test sets. Since the same subject may be included in several of the data partitions, the results of the analysis must be interpreted with this in mind.

The 5 approaches that can be used in this system are: 1) CNN1D, 2) LSTM, 3) BiLSTM, 4) BiLSTM-CNN, and 5) Logistic Regression. The different approaches can be used to model the relationship between the gait features and the different severity categories to be classified.

The models are trained with the training split of the data and the validation results are used to select the best model and also to select the best checkpoint for the model if required.

The results of the test trials are then compared with the true labels for the classes to be classified by the models. The accuracy, the balanced accuracy and the macroF1-score for the classification of the classes are then calculated in order to provide complementary results for the classification performance for the classes to be classified.

The metrics are compared for all models and data-preparation methods and interpreted in the context of the experimental setup and its limitations.

6.4 Proposed Flowchart

The workflow of the classification process is shown in Fig. 6. The process starts with the exploration and the validation of the source dataset, followed by the integration and the preprocessing of the movement data, and ends with the split of the prepared trials into the training set, the validation set, and the test set.

The chosen model is then trained with the training data, and the validation partition is then used to test the model. The chosen checkpoint is then tested with the test partition and the results for the classification metrics are saved. This process is then repeated for the remaining models and configurations.

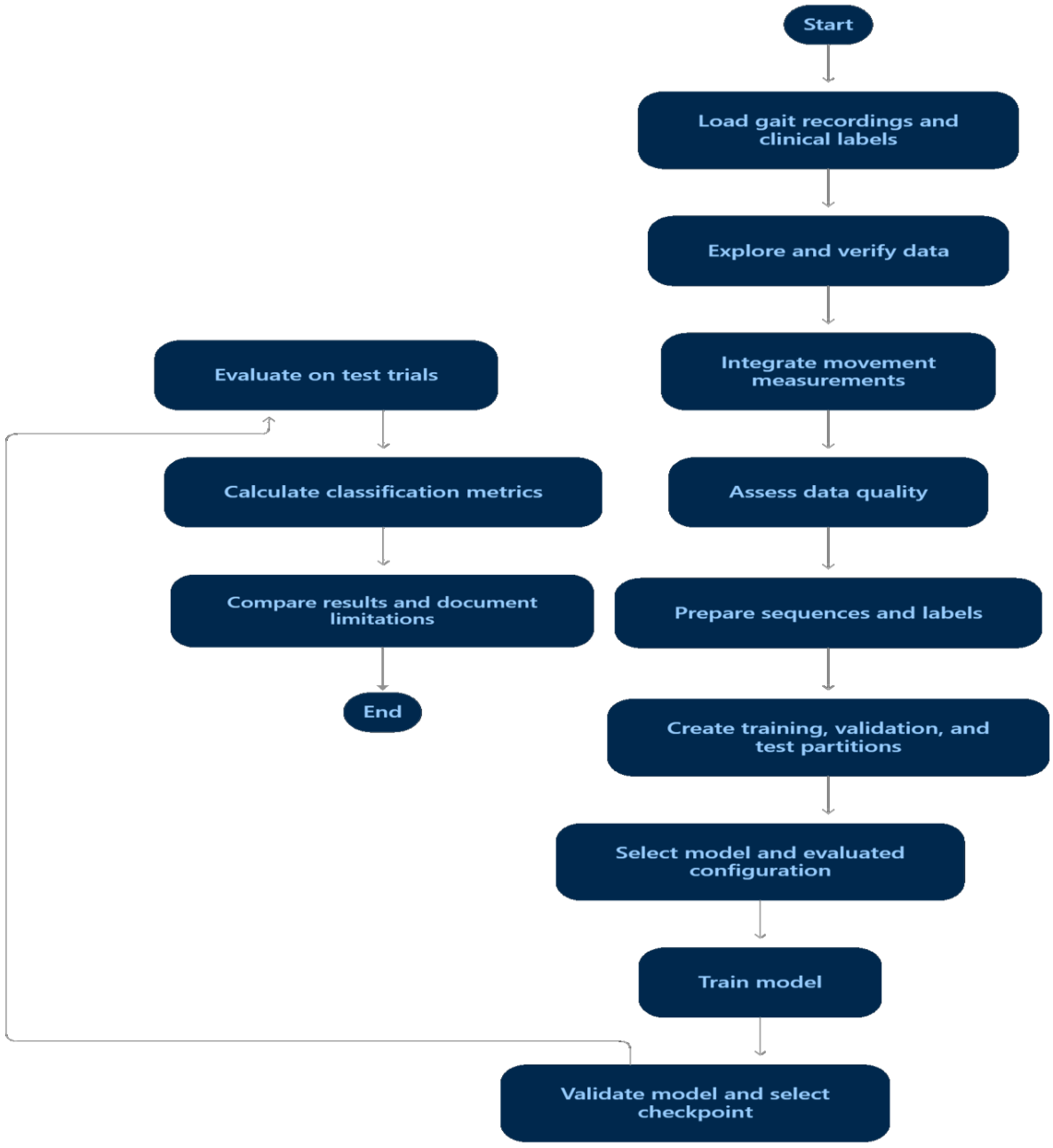


Fig. 6: Proposed classification algorithm flowchart

CHAPTER-7
IMPLEMENTATION (TOOLS AND TECHNOLOGY USED)

7.1 Introduction

The code for running the project in a Python environment for data exploration, sequential gait data preparation, machine learning model training, and performance evaluation. The implementation is divided into several stages that allow you to verify the data, train models, analyze the results step by step.

The use of technologies for numerical calculations, data analysis, deep learning, and traditional machine learning, as well as the storage of results from the experiment, are described. The output files are organized in a way that allows for stable evaluation of the models.

7.2 Tools and Technologies Used

7.2.1 Python

Python is the programming language used for the development of all the models described in this project. The Python programming language is highly supportive for data processing as well as for the development and evaluation of models.

7.2.2 Jupyter Notebook

Jupyter Notebook. Various notebooks are dedicated to data exploration, recording identification, association between movement and data, quality of data, preparation of sequences and evaluation of the model.

It allows easier inspection of intermediate results, checking out the properties of the data, and the experimental process.

7.2.3 PyTorch

The deep learning models are implemented and trained using the PyTorch [24] framework. This is a machine learning library that provides a tensor-based API, neural network modules, automatic differentiation, and a complete training framework.

The implementation of the models for the deep learning-based approaches is based on the PyTorch library. The deep learning approaches utilize the CNN1D, LSTM, BiLSTM and the hybrid model- BiLSTM-CNN as core frameworks for the implementation. The implementation further supports the use of CUDA for the acceleration of the training, validation and check point information of the models on systems with available GPUs.

7.2.4 Scikit-learn

We use the standard Logistic Regression classifier from the scikit-learn library to create a baseline model for our experiments, as well as to compute the various classification performance metrics. The various machine learning algorithms and utility functions provided by the scikit-learn library have been used for the implementation of the models as well as for their evaluation [25].

The library is also used to get the measures of accuracy, balanced accuracy and F1-score which can be used to compare the approaches faced by the project [7].

7.2.5 NumPy and Pandas

NumPy is used for numerical computations with multi-dimensional arrays of identical data type elements. Pandas is used to organize records in tables, clean data, process manifests and save results of analyses.

NumPy and Pandas are used within the experimental chain to prepare, check and then process the input data.

7.2.6 Configuration and Data-Storage Utilities

Configuration files are used for the dataset, the experiments, and the model evaluation. All information regarding the recordings, the trial splits, the training history, the predictions, and the evaluation results are stored in structured files.

These utilities are meant to help manage experiments and to avoid manual changes to files and locations.

7.3 Summary of the Technology Stack

Table 2: Technology Stack

Technology	**Purpose in the project**
Python	Main programming language
Jupyter Notebook	Organisation and execution of experimental stages
PyTorch	Deep learning model implementation and training
Scikit-learn	Logistic Regression and classification metrics
NumPy	Numerical and array-based computation
Pandas	Tabular data processing and experimental result management
CUDA	GPU acceleration for compatible deep learning operations

7.4 Implementation Details

7.4.1 Dataset Exploration and Verification

First of all, the relevant files for each subject, trial, recording condition and clinical data have to be identified by the file inventories and the file metadata and then the movement recordings have to be mapped accordingly.

The data flow process was used to check for missing files, incorrect file size, wrong target labels etc. The flow process was used to verify the input to the model as well as to check for the correct use of the input system for the data source. The data source for the gait studies of patients with Parkinson's disease contains walking kinematic and kinetic data for the studies [4].

7.4.2 Movement-Data Integration

The relevant linear and angular kinematic data of all gait trials were combined into trial representations.

The final input to the neural network models consists of 180 kinematic features for 751 frames of data (including frame and time).

7.4.3 Data Preparation and Validation

The implementation consists of data-quality evaluation, missing-value handling and sequential input preparation. Several data-preparation profiles were used in order to evaluate the impact on the classification results.

Validation is performed for files, shapes, classes and trial-partition integrity. Updating and using of so-called manifests describing the relationship between targets, data shapes and their respective experiment partitions is also part of the implementation.

The exact details of the various preprocessing steps are confidential so I do not go into further detail here.

7.4.4 Model Implementation

Five classification approaches are implemented and evaluated.

- CNN1D: Learns localized patterns of sequential movement measurements. Convolutional neural networks learn patterns of structured data [22].
- LSTM: Learns temporal dependencies between the gait sequences using recurrent processing [13].

- BiLSTM: Leverages information in both directions of a sequence for the modeling of time relationships [14].
- BiLSTM-CNN: Integrates recurrent and convolutional approaches to explore agments of sequential information [5].
- Logistic Regression: Baseline machine learning classifier for the dataset [21].

The different deep learning models as well as the Logistic Regression model are implemented using the PyTorch modules and the scikit-learn library respectively. All models are tested on the same comprehensive classification task.

7.4.5 Model Training and Validation

The last workflow uses a trial-wise split with 619, 131 and 131 training, validation and test trials respectively. To ensure reproducibility, a fixed seed for randomization has been used in the implementation.

The deep learning models are trained using their specific optimizers. The best model and its associated checkpoint are selected on the basis of the validation performance and other parameters as set up in the implementation. The training histories and the saved checkpoints can then be used to analyze the results of the experiments.

The same subjects are used for training, validation and test, so the results on the test partition do not provide proof of generalisation to new subjects. The design of the evaluation and the data leakage can also be problematic [8], [16].

7.4.6 Performance Evaluation and Result Storage

Once the selected models have been trained, they are used to make predictions for the test trial. These predictions are then compared to the true label for the test trial to determine the accuracy, the balanced accuracy, and the macro f1 score for the model [7].

"Accuracy" is simply an overall measure of correctness of prediction. Other classification measures " balanced accuracy" and others – give additional information about success of prediction for the target classes [7], [23].

The implementation saves the computed results such as metrics, predictions and additional training information into structured output files. This allows for an easy comparison of different models and experiments and is the basis for the results presented in the following chapter.

Table 3: Phases and corresponding notebooks

Phase	**Notebooks**
Dataset exploration and understanding	1-2
Data collection,organization and preparation	3-5
Feature extraction and feature engineering	6-7
Data normalization and scaling	8
Sequence windowing and label preparation	9
Data quality and preprocessing validation	10
Baseline model development and performance evaluation	11

7.5 Implementation Summary

The system is a pipeline of Python data-processing tools, PyTorch deep learning models, and scikit-learn classification tools structured as an experimental process. Separate sections are devoted to data validation, sequential processing, training of models, validation of predictions, and output of results.

The system was used to conduct a comparative study between classic machine learning approaches and deep learning approaches for the three-class classification of Parkinsonian gait severity. However, the results are restricted by the small sample size and the fact that the results were obtained using a trial-wise evaluation, meaning that the results do not necessarily generalize to novel subjects.

CHAPTER-8
RESULT DISCUSSION AND ANALYSIS

8.1 Introduction

This chapter is about the experimental evaluation of machine learning and deep learning models for three-class classification of Parkinsonian gait severity. An evaluation of CNN1D, LSTM, BiLSTM, the hybrid BiLSTM-CNN model architecture and Logistic Regression as a standard baseline.

The experiments test model performances with several data-preparation setups. Classification performances are compared by accuracy, balanced accuracy, and macro F1-score. Values in this chapter are obtained from the final-results notebook labelled on the appendix [7].

8.2 Experimental Setup

Final test: Gait sequences with 751 time steps and 180 kinematic features were used in the final test. The label is the walk-related UPDRS-III, that is grouped into three classes with 0, 1 and 2.

Table 4: Trial-wise dataset partition

Dataset Partition	Number of Trials	Percentage
Training	619	70.41%
Validation	131	14.90%
Testing	131	14.90%
Total	881	100%

Any small variations from 70/15/15 are due to incorporating whole trials into each partition.

The five models were tested in four data-preparation settings, for 20 total model-data-preparation combinations. The final notebook contains valid evaluation results for all 20 combinations.

One notable feature of the experimental design is that trial id's are distinct across partitions, but not subjects. All 26 subjects are present across all the training, validation and test partitions. We take this into account when discussing the performance reported.

8.3 Evaluation Metrics

The results of the classification are evaluated using three metrics.

Accuracy: Indicates the ratio of the number of correct test predictions to the actual class. An estimate of the overall correctness of classification.

Balanced accuracy: Averages recall for the three classes so that each class gets the same weight in the overall score. Useful for evaluating performance when class distributions are unequal.

Macro F1-score: Computes the F1-score for each class individually, then takes the unweighted mean of the three scores. It is a useful metric when you need to balance precision and recall.

These parameters also give an overall evaluation that accuracy can not.

8.4 Comparative Model Performance

Table 5: Best observed test performance for each model

Model	Best Accuracy	Best balanced accuracy	Best macro F1-score
CNN1D	0.984733	0.978495	0.981481
LSTM	0.954198	0.941838	0.950366
BiLSTM	0.984733	0.978495	0.983041
BiLSTM_CNN	0.984733	0.978896	0.983041
Logistic Regression	0.984733	0.978495	0.981481

As with the different feature extraction methods, CNN1D, BiLSTM, BiLSTM-CNN, and Logistic Regression achieved the same highest test accuracy of 98.47%. The highest value for the balanced accuracy metric was 97.89%, by the hybrid BiLSTM-CNN model for one of the configurations tested, and the maximum value for macro F1-score was 98.30%, achieved by both BiLSTM and BiLSTM-CNN for their respective configurations.

The max accuracy, balanced accuracy, and macro F1-score obtained by the LSTM model were 95.42%, 94.18% and 95.04% respectively. The maximum values reported here were worse than the best maximum values reported by any of the remaining techniques.

8.5 Analysis of Individual Models

8.5.1 CNN1D

CNN1D obtained the best test accuracy of 98.47%, balanced accuracy of 97.85% and macro F1-score of 98.15%. These results imply that the convolutional processing is able to find meaningful cues in the proposed gait sequences while using the current evaluation framework. The classification matrix of model is shown in Fig. 7.

But the scores differed, depending on which configuration of data-preparation was used. That shows that the performance reported is not a property of the model alone, but also of the way the input data is prepared.

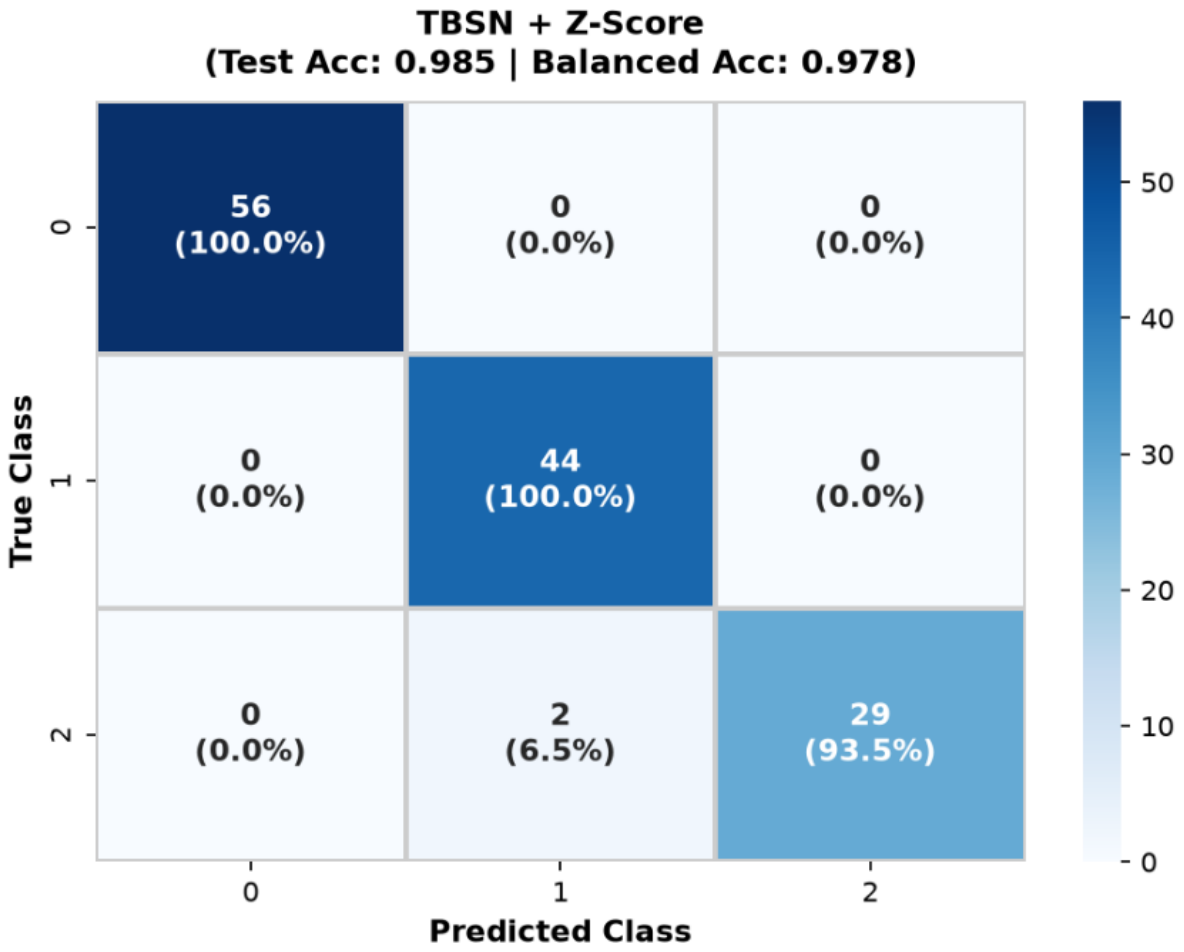


Fig. 7: Classification Matrix of CNN1D

8.5.2 LSTM

The LSTM model attained a maximum test accuracy of 95.42% and a maximum macro F1-score of 95.04%. Despite not outperforming many of the other models with the tested configuration, the LSTM model gives an interesting point of comparison when studying recurrent sequence modelling. The classification matrix of model is shown in Fig. 8.

The findings also show why we cannot assume that a model for temporally structured data, will always outperform other approaches, and that we need to assess models by making direct comparisons.

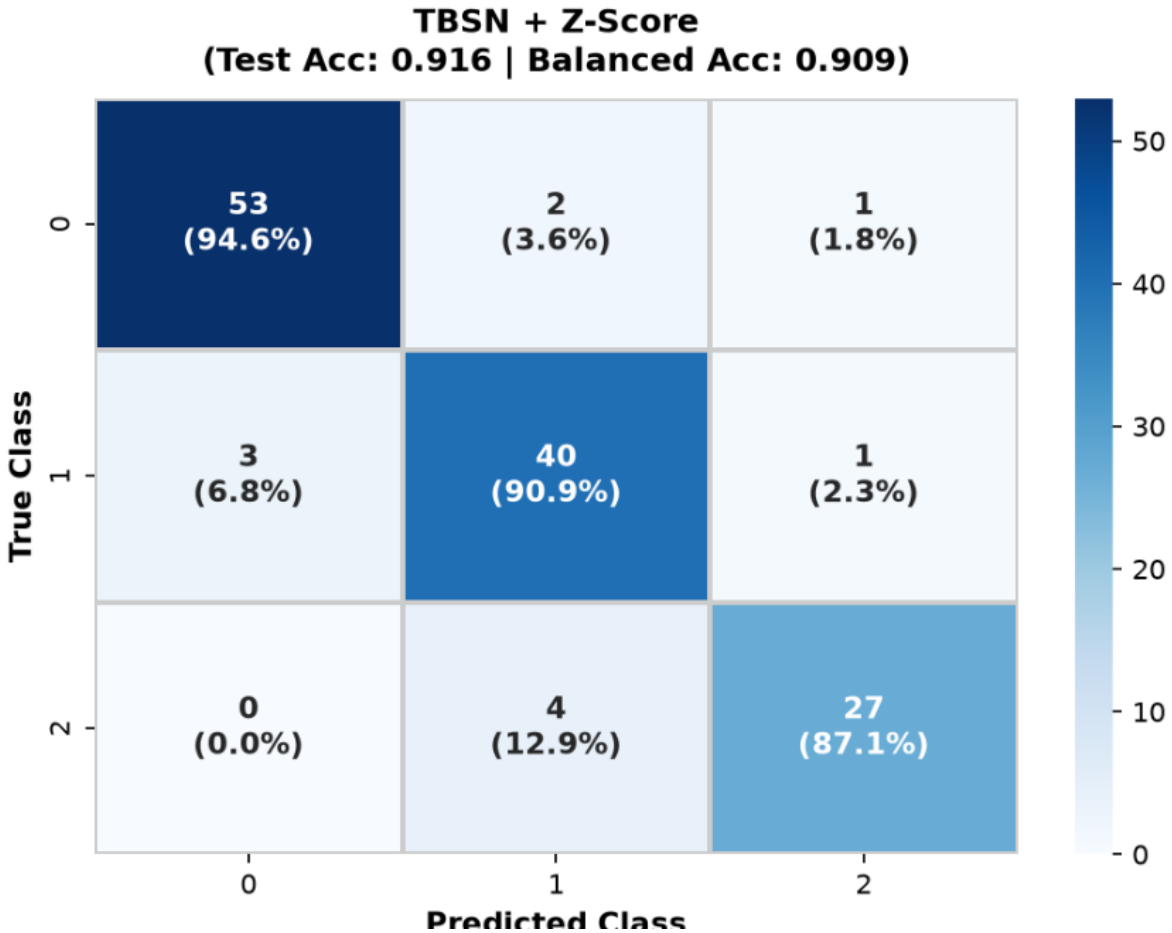


Fig. 8: Classification Matrix of LSTM

8.5.3 BiLSTM

The BiLSTM model attained a maximum test accuracy of 98.47%, a balanced accuracy of 97.85%, and a macro F1-score of 98.30%. Its highest macro F1-score was equal to that of the hybrid model. The classification matrix of model is shown in Fig. 9.

These results suggest that bidirectional temporal modelling is a promising approach for the selected classification task. Nevertheless, the result should be interpreted in the context of the trial-wise evaluation rather than as confirmed generalization to unseen individuals.

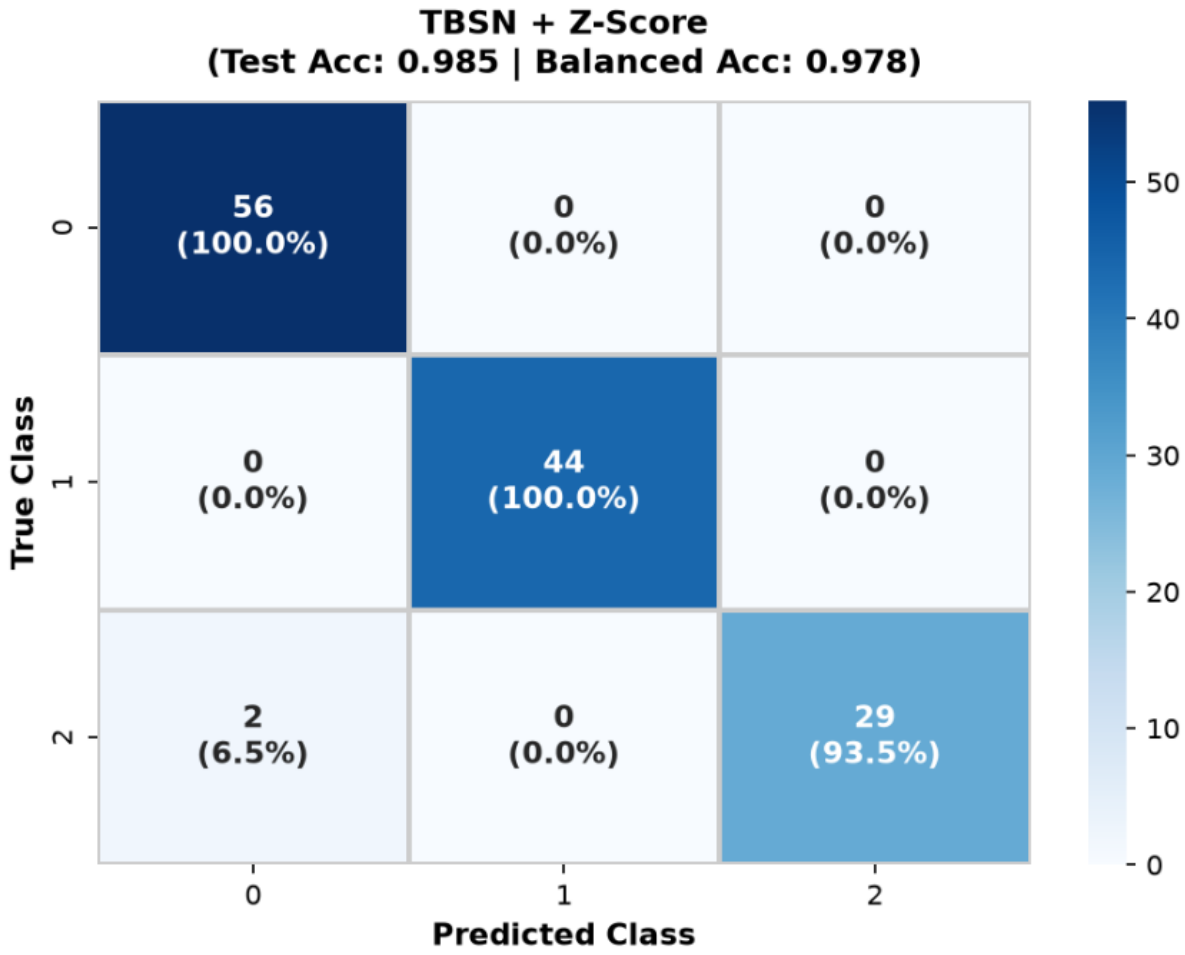


Fig. 9: Classification Matrix of BiLSTM

8.5.4 Hybrid BiLSTM-CNN

The hybrid BiLSTM-CNN architecture obtained a maximum test accuracy of 98.47%, maximum balanced accuracy of 97.89%, and maximum macro F1-score of 98.30%. The classification matrix of model is shown in Fig. 11.

Its balanced accuracy was slightly above the maximum for the other models. Its macro F1-score was also equivalent to that of BiLSTM. This hybrid architecture is therefore a potential model for additional investigation.

The difference of the best observed scores is small and the maxima can be achieved by different configurations. Therefore, these results on their own do not confirm that the hybrid architecture is always preferable in all data-preparation settings.

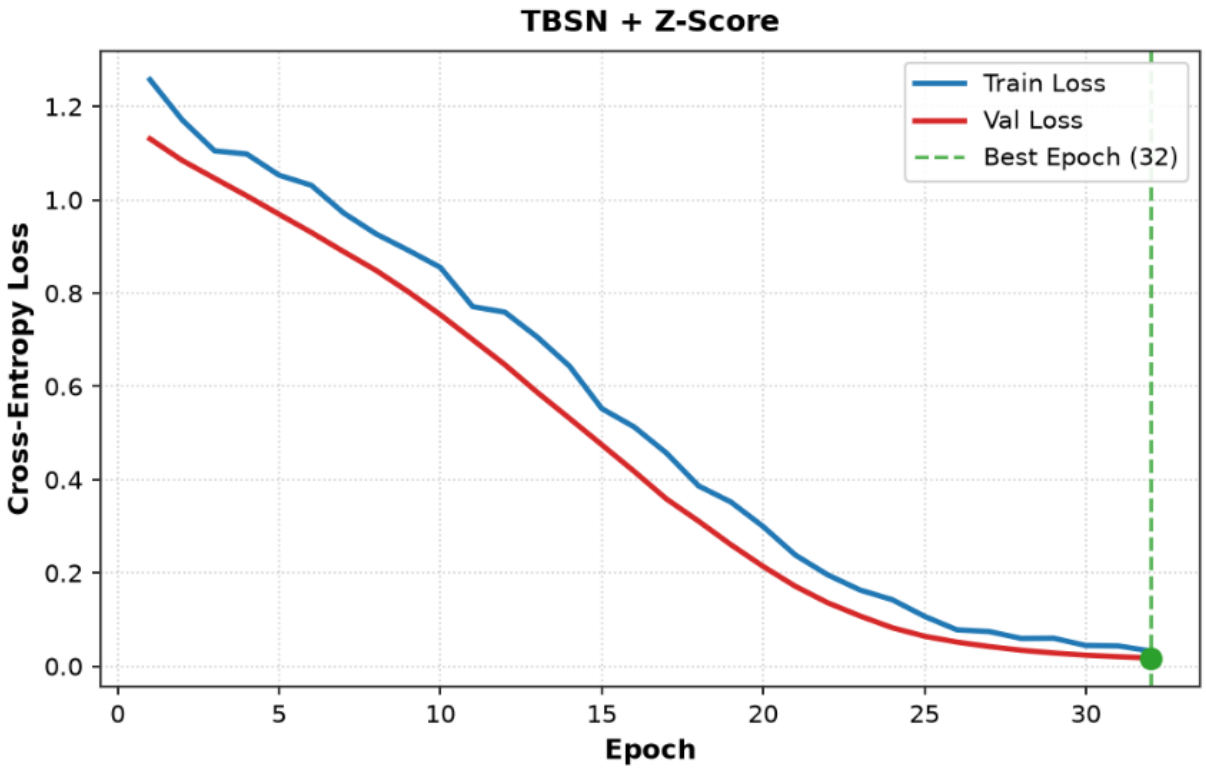


Fig. 10: Loss Vs Epoch

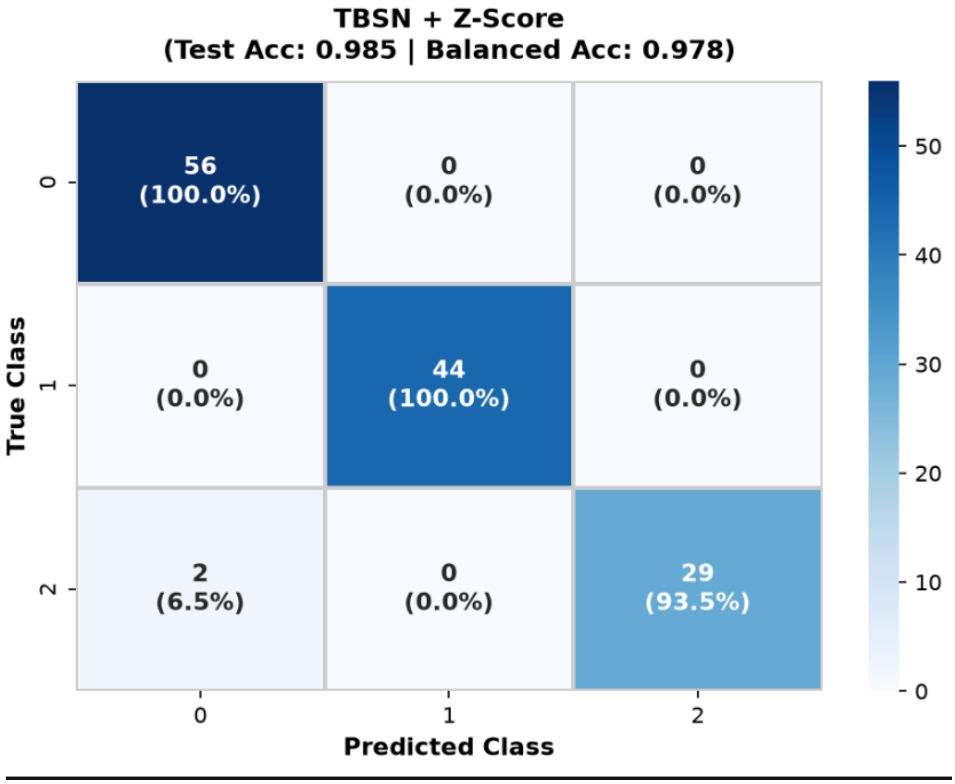


Fig. 11: Classification Matrix of BiLSTM_CNN

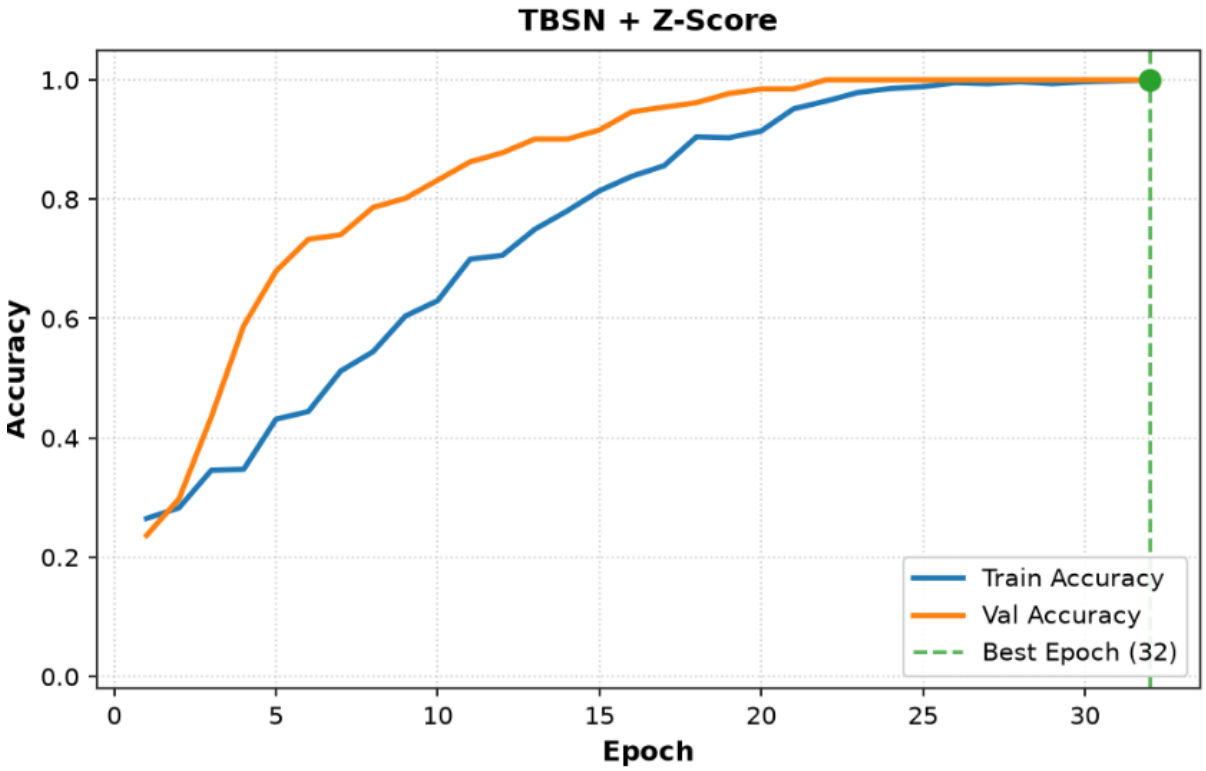


Fig. 12: Accuracy Vs Epoch

8.5.5 Logistic Regression

Conventional baseline Logistic Regression showed a maximum test accuracy of 98.47%, a maximum balanced accuracy of 97.85% and a maximum macro F1-score of 98.15%. The classification matrix of model is shown in Fig. 13.

Its accuracy is remarkable, in the fact that it achieved the same maximum accuracy as multiple deep learning models on the experiments that were tested. This shows that, under the experimental conditions, a more simple model can serve as a reference on this task.

This outcome does not prove that Logistic Regression is, in general, equivalent to the deep learning architectures. A more general, subject-independent test would need to be performed to show how the methods compare on unseen subjects.

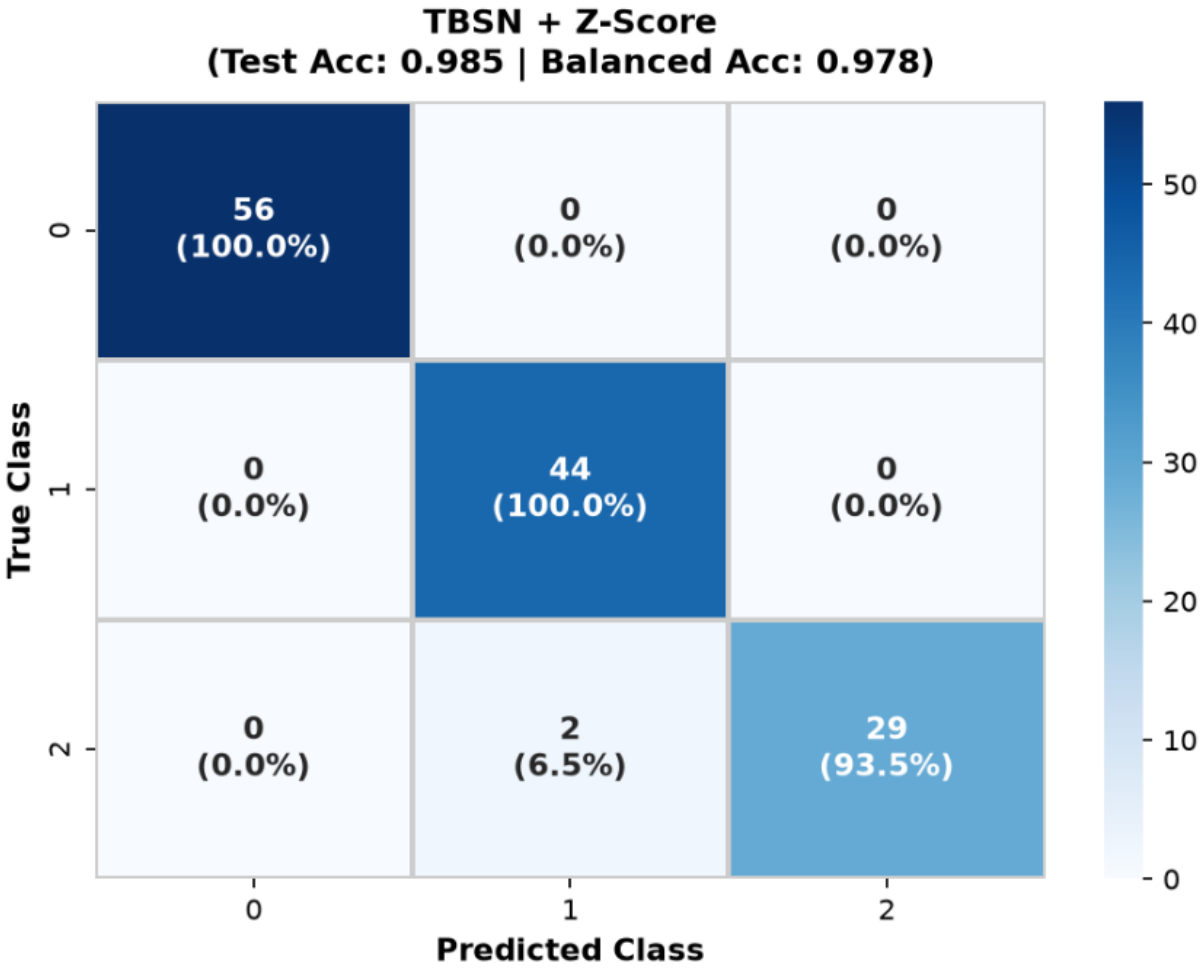


Fig. 13: Classification Matrix of Logistic Regression model

8.6 Comparative Interpretation

From the comparison we can extract the following three observations.

Firstly, there was a number of models that obtained the same maximum test accuracy of 98.47%. Accuracy alone is thus not enough to clearly differentiate between the more effective methods. Balanced accuracy and macro F1-score give us more details about their performance on the three target classes.

Secondly, the combined architectures of hybrid BiLSTM-CNN and BiLSTM achieved the best balanced accuracy of 97.89% and the best macro F1-score of 98.30% respectively, with the latter combined architecture achieving the best macro F1-score. They are both promising architectures to further investigate, but the marginal differences failed to find a clear winning model.

Third, performance differed depending on data-preparation schemes. This emphasizes the advantage of selecting a model by evaluating the entire experiment pipeline, rather than merely the architecture.

In general, there is high scoring with the used trial-wise split. But the following scoring should be considered as a measurement of experimental performance in the used evaluation setting, not an estimate of performance in an entirely independent population.

8.7 Limitations of the Results

The findings should be understood within the context of the following limitations.

The dataset comprises recordings from 26 subjects. While different multiple trials facilitate model evaluation, these did not represent an equal number of independent participants.

Split in trial-wise subjects: Subjects are present in more than one partition, including the test set. Models will therefore be trained on some subject specific movement characteristics that are also present in test recordings; this may make the results less representative of unseen subjects.

Configuration-dependent performance: The highest observed value for one metric may be achieved under a different configuration of data-preparation parameters than the highest observed value for a different metric. The maximum values reported in Table 5 should be interpreted as metric-wise maxima, not the maximum from one configuration.

Requires additional validation:Results on the subject-independent partition and other recordings should be investigated to see if the results were also achieved on participants who did not take part in the training.

However, these limitations do not dismiss the results that have been recorded. They merely illustrate the extent of conclusions that can be appropriately made based on the current experiments.

8.8 Summary

The experimental evaluation of five classification techniques under four data-preparation settings resulted in the following performance measures: a maximum test accuracy of 98.47%, maximum balanced accuracy of 97.89%, and maximum macro F1-score of 98.30%.

CNN1D, BiLSTM, BiLSTM-CNN and Logistic regression were reached in the measuring experiments, with BiLSTM-CNN obtaining the highest balanced accuracy and both BiLSTM-CNN and BiLSTM achieving the highest macro F1-score. LSTM had lower maximum scores in the tests performed.

The results show encouraging classification accuracy within the performed experiment. Due to the trial-wise split with subject overlap between splits, additional subject-independent tests must be performed before any conclusions can be made regarding generalization to new subjects or a clinical setting.

CHAPTER-9
CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

In this project, both machine learning and deep learning approaches were used for the classification of Parkinsonian gait severity based on kinematic sensor data into three classes. The workflow of this project consists of three main parts, namely dataset exploration, integration of movement-data, data preparation as well as the development of classification models and comparison of results.

Five different classifiers (CNN1D, LSTM, BiLSTM, the hybrid BiLSTM-CNN model and the classical Logistic Regression baseline approach) are compared in terms of their classification performance for several different versions of the data-preparation. Accuracy, balanced accuracy and the macro average F1-score are shown for each of the classifiers.

From the results it is possible to identify computationally some of the features of gait that can be used to classify its severity into three classes. The results obtained with the BiLSTM models and the hybrid BiLSTM-CNN model are very good. The results obtained with the Logistic Regression baseline model are also very good.

The results of the experiments in this study were different for each of the classification setups. Therefore, both the model and the data preparation used for classification had an influence on the results of the classification. To make a more detailed evaluation of the classification performance over the target classes several measures for classification performance were used [7], [23].

A major limitation of the above-mentioned evaluation method in our experiments was the trial-wise split into training, validation and test set. Therefore, recordings from the same subjects are in some cases used in the training, validation and test set. Therefore, the obtained scores are valid for this specific setup only and do not have to hold for completely unseen subjects [8].

This work serves as a reference for comparison of current approaches of Parkinsonian gait severity classification based on sensors. It can serve as a starting point for further work and of course needs to be tested on other approaches before general or clinical application is possible.

9.2 Future Scope

These ideas can be developed further.

1. Partitioning the future data into training, validation and test sets would allow for a more precise assessment of the subject independence of the classifiers [16] (i.e. the classification

error on subjects not used for training). The validation must be done very carefully when data is pooled by subject or otherwise correlated [8].

2. More subjects and more gait recordings should be included in the experiments. Also more performance variables and more recording conditions with a more diverse dataset should be tested.

3. For future studies, different sequence-learning architectures could be compared to the models presented here within the same experimental setup. Model selection should be based on several metrics instead of solely on accuracy [7].

4. It would also be useful to run the same experiments on different random initializations with a validation set for model selection, in order to get an idea of the stability of the results and maybe more information about the performance.

5. Model interpretability: We can investigate the most important movement features used by the models to make predictions and interpret the results for analytical purposes. However, such results need to be validated carefully before they can be used for clinical purposes.

6. It would be of interest to test the performance of the classification method across different medication conditions, recording sessions and even subjects. Such a study would be valuable for wearable-sensor based gait assessment as it allows to test the correspondence between the measured movement features and the disease severity across different conditions [19].

7. The method will be tested independently and then can be used as an auxiliary tool in addition to the clinical assessment. For real-world gait analysis, studies on sensor-based gait assessment are needed to test the approach in an uncontrolled environment [20].

In summary, subject-independent testing, more data and more validation is needed for future studies using the proposed approach for sensor-based gait-severity classification.

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Github Repository:

https://github.com/ATUL-IIITB/GROUP21_MINOR_PROJECT